

Portfolio: github.com/Leo-Marcinkus/Engineering-Portfolio/

Education:

University of Colorado Boulder, Boulder, CO

B.S. Electrical Engineering - College of Engineering and Applied Science – Expected May 2027

Accelerated M.S. Electrical Engineering – College of Engineering and Applied Science – Expected May 2028

Interests/Hobbies: Audio electronics and production, Mechanical watches, Guitar, Piano, NBA, benchmarking components, Jimmy John’s, EDC

Projects:

Golden Arduino Board: Designed and built a custom high-performance Arduino board. Used Altium, interpreted and implemented multiple data sheets, and analyzed similar microcontrollers to design and assemble a custom Arduino with multiple design improvements.

Retro Glass Fuse Lamp: Created a retro aesthetic small scale lamp utilizing glass fuse casings, LED Edison bulbs, and an integrated rechargeable battery circuit. Improved upon the original inspiration by implementing ESP-Now control, improvements regarding power efficiency, and good practices in battery safety.

Alternative Radio Tuning: Designing a radio receiver utilizing an alternative capacitor-switching LC tuning strategy and a software-controlled frequency selection across the AM/FM broadcast bands. I implemented analog switching and audio amplification to receive real-time broadcasts and validate tuning using the NIST 60 kHz reference signal. The prototype involved system involved various breadboard circuits before being developed into a custom PCB for lower noise and better RF performance.

Apartment Rover: Designed and built a robot to autonomously navigate my apartment. Utilized an Arduino with a proximity sensor and created the speed sensor, direction control, and speed regulation from scratch on a breadboard. Incorporated an OLED display for data visualization.

Sumo Robot: Developed a basic sumo robot to learn embedded systems. It primarily served to introduce embedded systems and MicroPython. Gained hands-on experience with MicroPython, motor drivers, and IR/RF receivers.

Dynacone: Engineered a dynamically moving sports cone programmed to execute diverse training patterns. Utilized an Arduino, prototype PCB, motors, and a servo for precise movement.

Lab and Circuit Experience:

Advanced Electronic and Semiconductor Laboratory August 2025 – December 2025

- Analyzed active and passive semiconductor devices (BJTs, diodes, photodiodes, MOSFETs) using oscilloscopes, DMMs, waveform generators, and various power supplies
- Performed I-V measurements, parameter extraction, and curve fitting to evaluate device performance and variability using industry-standard software
- Analyzed experimental results and correlated data with theoretical models in formal technical reports

PCB Design and Manufacturing August 2025 – December 2025

- Designed multi-layer PCBs in Altium from schematic design to fabrication-ready layouts
- Utilized oscilloscopes to conduct signal integrity, crosstalk, and power-rail noise measurements
- Applied PDN and layout best practices through breadboard prototyping and PCB implementation

Embedded Software and Real-Time Operating Systems August 2025 – April 2026

- Developed embedded firmware on ARM Cortex-M microcontrollers using C, registers, and memory-mapped peripherals
- Designed multi-threaded real-time applications using FreeRTOS with semaphores, mutexes, and task scheduling
- Implemented debugging, tracing, and profiling to validate timing and ensure optimal system performance

Electronics Design Lab August 2024 – December 2024

- Modeled DC motors and implemented PWM-controlled H-bridges with PID feedback for real-time velocity control using an Arduino
- Designed and tested active op-amp low-pass filters and encoder interface circuits for optimal signal conditioning
- Performed DC, transient, and frequency domain analysis in the SPICE-based software SIMetrix to optimize and debug circuit performance

Skills:

Hardware and PCB

- PCB design, multi-layer layouts, schematic testing, Altium, signal integrity, PDN design, power management

Electronic Analysis

- Analog circuit design, circuit testing and debugging, I-V characterization, LTSpice, ADS, HyperLynx

Embedded Systems

- STM32 boards, Arduino, Microcontrollers, PWM, SPI/I²C/UART, Real-Time Systems, RISC-V

Programming Tools and Systems

- C++/C, Python/MicroPython, SystemVerilog, Assembly, CAD

Testing Tools and Verification

- Oscilloscope, logic analyzer, DMM, data validation, waveform generators

Work Experience:

Lanzi Designs and Cabinetry – Independent Contractor December 2022 – January 2026

- Ran material pickup and delivery to ensure efficiency of project completion
- Measured, cut, and assembled cabinets with precision to meet custom specifications
- Installed cabinetry, flooring, trim, and ceiling support for structural integrity and aesthetics
- Professionally engaged with clients to understand needs and provide quality craftsmanship

Eddie Bauer – Sales Associate September 2022 – May 2023

- Operated the register, processed transactions, and assisted customers with purchases
- Resolved customer concerns and returns through clear communication and conflict management
- Provided personalized customer experiences and answered questions with product recommendations

Express – Sales Associate May 2022 – August 2022

- Replenished and organized inventory to provide a well-stocked floor
- Engaged with customers to provide styling advice and case by case recommendations
- Assisted in closing responsibilities to prepare the store for corporate check-in to meet with CEO and executives